Bilal Ahmad

Electrical Engineer | Power Systems Researcher | Machine Learning

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PROFILE

PhD researcher at the University of Bath working on power distribution network reliability and Battery Energy Storage Systems, combining Optimal Power Flow (OPF/SCOPF) with explainable machine learning for grid resilience. Former Technical Lead in Machine Learning at HCL Technologies, with a track record spanning power system optimisation, control engineering, and applied AI across research and industry. Currently also working as a full stack developer alongside the research.

PROFESSIONAL EXPERIENCE

Full Stack Developer — Dystil.AI — Apr 2026 – Present · Part-time, Remote
- Full-stack web development across front-end interfaces and back-end services
- Feature implementation, code review and release support

Full Stack Developer — Just Jutz — Apr 2026 – Present · Part-time, Remote
- End-to-end web development, from user interface through to server-side logic
- Building and maintaining features across the existing codebase

Teaching Assistant — University of Bath — Jan 2026 – Present
- Supported labs, tutorials and grading across computing and electrical modules

Teaching Assistant — University of London — Apr 2023 – Jun 2026
- Supported labs, tutorials and grading across computing and electrical modules
- Held office hours and provided one-to-one support for coursework and projects
- Coordinated assessment rubrics and feedback to improve learning outcomes

Visiting Researcher — Technical University of Denmark — May 2026

Research Assistant — Royal Holloway, University of London — Aug 2023 – Oct 2025
- Contributed to ML-driven power systems research and prototype development
- Managed datasets, experiments and result reproducibility pipelines
- Prepared figures, drafts and proofs-of-concept for publications

Technical Lead, Machine Learning — HCL Technologies — Aug 2022 – Jul 2023
- Led development of ML-enabled features across products, including handwritten text extraction, OCR, and automated catalogue tagging
- Built sales forecasting and document digitisation pipelines, and an image noise-removal model for MedTech-relevant sensor systems
- Designed a fully automated OPF and SCOPF framework applying ML-based power flow analysis for grid contingency screening

Visiting Researcher — RWTH Aachen University — Oct 2021 – Mar 2022
- MTDC microgrid model in MATLAB
- ADRC scheme for buck, boost and DAB converters
- Robustness and convergence analysis
- Small-signal, sensitivity and modal analysis

Intern — PricewaterhouseCoopers — Aug 2021 – Oct 2021
- Artificial intelligence in smart grids
- Advanced metering infrastructure
- Smart grid contracts

Intern — Harduaganj Thermal Power Station — May 2017 – Jun 2017
- Coal handling plant supervision
- Cooling unit and switchgear supervision
- SCADA monitoring and workforce management

EDUCATION

PhD, Power System Reliability — University of Bath — Current
- Researching power distribution network preparedness for severe weather events involving compound hazards
- Developing an automated optimal procurement strategy framework for Distribution Network Operators (DNOs)
- Applying explainable AI (XAI) to intelligent contingency screening for power system reliability

MPhil, Power System Optimisation using Machine Learning — Royal Holloway, University of London

M.Tech, Electrical Power Systems — IIT Roorkee Distinction

B.Tech, Electrical Engineering — Aligarh Muslim University

PUBLICATIONS

- B. Ahmad and O. Nduka, "A Comparative Analysis of Machine Learning Based Power Flow Study with Custom Made Open Source Python Codes," IEEE icSmartGrid 2025.
- B. Ahmad, S. M. Amrr, M. Nabi, M. R. Khalid and M. S. J. Asghar, "Analysis of Three-Phase Grid-Tied Thyristor Based Inverter for Solar PV Applications," IEEE SEFET 2021.
- M. R. Khalid, B. Ahmad, A. Sarwar and M. S. J. Asghar, "Review of Thyristor Based Grid Tied Inverters for Solar PV Applications," IEEE I2CT Pune 2019.
- Forthcoming — International Journal of Renewable Energy Research.

KEY PROJECTS

- Power System Reliability in Adverse Weather Events — Bath PhD research programme
- XAI-based SCOPF Framework for Power System Optimisation — RHUL MPhil programme
- System-Level Control of Inverters in DC Microgrids — RWTH Aachen thesis
- Multilevel Inverter with MPP Tracking — power electronics research project

TECHNICAL SKILLS

Languages Python · MATLAB · Java · JavaScript · HTML · CSS
Optimisation and power systems Pyomo · PandaPower · MATPOWER · Gurobi · Ipopt
Data and machine learning Pandas · NumPy · SciPy · Matplotlib

FELLOWSHIPS AND AWARDS

Fellowships URSA · DAAD · LSC · MHRD
Award Third Best Paper — icSmartGrid 2025, Glasgow, UK